8.2 Integration by Parts.

This is derived from the product rule that is used to find the derivative of two products. Such as;

$$\frac{d}{dx}[uv] = u \cdot \frac{dv}{dx} + \frac{du}{dx} \cdot u$$

$$uv = \int udv + \int vdu \qquad \text{Applying integration.}$$

$$\int udv = uv - \int vdu \qquad \text{Rearranging}$$

Note that there are certain ways to make the formula work;

1. Peak dv to be the most complicated part of the integral and must contain dx.
2. Let u to be part of the integral that has a simpler derivative du.

   It is important that there are various resources to aid in your learning of calculus that can be used determine the answer to a solution is accurate. Midpoint rule is one such which you can utilized computer applications such as excel, matlab and Mathematica to determine approximately whether you answer is correct. The one set back though is the accuracy of the answer computed may not be given.

- When carrying out integration by parts be sure not to interchange the substitution. This will end up undoing the initial integration and leading you to the original problem. Also, watch out for the repeated constants which can be moved to the other side of the equation thus whatever, is obtained has to be divided by a multiple.